

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2022-23

Paper: BVCA-303 — Programming in Linux

Time: Three Hours

Maximum Marks: 70

Instructions: Answer all Sections. Candidates are required to write their answers in their own words as far as practicable.

SECTION — A: Long Answer Questions

[15 × 2 = 30]

Instructions: Long answer type questions to be answered in about 750 words each.

Question 1.

- (a) What is an Operating System? Explain the Linux Operating System architecture and list its key features.
- (b) What are **inode** and **dentry** in the context of the Linux file system? Describe the Linux file system structure in detail.

OR**Question 2.**

- (a) What is octal permission notation? Discuss the Linux file permission model with suitable examples.
- (b) What is context switching? Explain the various states of a Linux process in detail.

SECTION — B: Short Answer Questions

[5 × 6 = 30]

Instructions: Attempt any six questions. Short answer type questions to be answered in about 550 words each.

Question 3. Write a shell program to calculate the average and sum of a given set of numbers.

Question 4. Write a shell program to compute the factorial of a given number.

Question 5. List and describe the different modes of the **vi** editor.

Question 6. Explain the working and usage of the **awk** utility with suitable examples.

Question 7. Explain what background processes are and how they are managed in Linux.

Question 8. Discuss any five essential Linux commands with examples.

Question 9. Write a brief note on the Linux shell and its different types.

Question 10. Discuss briefly the Dash, Activities, and Grid GUI elements in Ubuntu Linux.

SECTION — C: Very Short Answer Questions

[10 × 1 = 10]

Instructions: Answer the following questions in a single word or sentence.

1. What is the command used to display the MAC address of a Linux machine?
2. Which command is used to rename a file in Linux?
3. Write the command used to print a file from the terminal.
4. In which system file are user passwords saved in Linux?
5. What option of **grep** enables us to ignore the case of alphabetical characters?
6. Name the boot loader commonly used in Ubuntu Linux.
7. Which command is used to search forward in the **vi** editor?
8. Name the processes that run unattended in the background.
9. What key combination will terminate a running command in the terminal?
10. Name the software that runs the GUI environment in Ubuntu Linux.